The goal of this assignment is to develop an agent that will allow the remote execution of tasks. Pay attention:

- The code should be written in production quality.
- Design and interfaces are more important than fully implemented functionality.

Part A:

Requirements:

- The agent should support remote API for "run task" commands

Example for tasks:

Task	**Parameters**	**Return value**
DNS query	• Domain	Resolved IP address.
Preform HTTP request	• Method • Domain / IP • Port • Path	Body returned from the request.
Port scan	• Domain / IP • From port • To port	List of open ports.
Process tree	• PID	Process tree from the root PID.
Registry	• Action (GET / SET / DELETE) • Key • Value (optional)	Success/Fail otherwise.
Net Share	• Action (GET / ADD / DELETE) • Name (optional)	Success/Fail otherwise.
Start an HTTP server	• Port • Page URI • Data to return when accessing the page	Who requested the page.

- Tasks' execution may fail
- The agent should be extendable to new tasks
- Tasks should run in parallel

Part B (Optional):

- Write a test that will involve 2 agents

Good Luck!